

EA-0024 - AQELYN Engineering Archive

AQELYN - EA-0024 Engineering Archive

IS-024 - AQELYN Vulnerability Intelligence & Prioritization Engine

****Archive ID:**** EA-0024

****Implementation Specification:**** IS-024

****Component:**** AQELYN Vulnerability Intelligence & Prioritization Engine

****Project:**** AQELYN

****System Type:**** Cyber Security Operating Environment

****Status:**** COMPLETE

****Repository Impact:**** No top-level repository structure changes

****Breaking Changes:**** None

****Engineering Phase:**** Phase 3

****Predecessor Archives:**** EA-0001 through EA-0023

****Next Specification:**** IS-025 - AQELYN Cyber Asset Discovery & Inventory Intelligence Engine

Document Control

| Field | Value |

---|---

| Document | Engineering Archive EA-0024 |

| Specification | IS-024 - AQELYN Vulnerability Intelligence & Prioritization Engine |

| Publication Format | Markdown, PDF, HTML, ZIP |

| Source of Truth | MD/EA-0024.md |

| Archive Rule | Implementation Specification -> Engineering Archive -> Continue |

| Repository Rule | Fixed repository structure; no redesign |

| Completion State | IS-024 complete; EA-0024 generated |

1. Engineering Context

AQELYN is being built as a modular Cyber Security Operating Environment.

Completed engineering archive chain:

| Engineering Archive | Implementation Specification | Component |

---|---|---

| EA-0001 | IS-001 | AQELYN Kernel |

| EA-0002 | IS-002 | Universal Object Model |

| EA-0003 | IS-003 | AQELYN Event Bus |

EA-0004	IS-004	AQELYN Evidence Engine
EA-0005	IS-005	AQELYN Knowledge Graph
EA-0006	IS-006	AQELYN Trust Engine
EA-0007	IS-007	AQELYN Mission Engine
EA-0008	IS-008	AQELYN Workflow Engine
EA-0009	IS-009	AQELYN Policy Engine
EA-0010	IS-010	AQELYN Compliance & Governance Engine
EA-0011	IS-011	AQELYN Identity & Access Governance Engine
EA-0012	IS-012	AQELYN Asset & Configuration Governance Engine
EA-0013	IS-013	AQELYN Risk Intelligence Engine
EA-0014	IS-014	AQELYN Threat Intelligence Fusion Engine
EA-0015	IS-015	AQELYN Security Operations (SOC) Engine
EA-0016	IS-016	AQELYN Digital Forensics Engine
EA-0017	IS-017	AQELYN Threat Detection & Analytics Engine
EA-0018	IS-018	AQELYN Automated Response & Orchestration Engine
EA-0019	IS-019	AQELYN Security Data Lake & Telemetry Platform
EA-0020	IS-020	AQELYN AI Decision Intelligence Engine
EA-0021	IS-021	AQELYN Predictive Analytics & Forecasting Engine
EA-0022	IS-022	AQELYN Executive Intelligence & Strategic Reporting Engine
EA-0023	IS-023	AQELYN Threat Exposure & Attack Surface Management Engine
EA-0024	IS-024	AQELYN Vulnerability Intelligence & Prioritization Engine

The fixed repository structure remains:

```
AQELYN/  
■■■■ archive/  
■■■■ blueprint/  
■■■■ docs/  
■■■■ src/  
■■■■ tests/  
■■■■ tools/  
■■■■ build/  
■■■■ releases/  
■■■■ scripts/  
■■■■ assets/  
■■■■ examples/  
■■■■ plugins/  
■■■■ sdk/  
■■■■ api/  
■■■■ README.md
```

The engineering rule remains:

```
Finish Implementation Specification  
↓  
Generate Engineering Archive  
↓  
Continue
```

No Engineering Archive may be skipped.

2. IS-024 Specification Identity

```
Specification ID: IS-024  
Name: AQELYN Vulnerability Intelligence & Prioritization Engine  
Engineering Archive Target: EA-0024  
Project: AQELYN  
System Type: Cyber Security Operating Environment  
Status: Complete  
Predecessor: IS-023 - AQELYN Threat Exposure & Attack Surface Management Engine
```

3. Purpose

The AQELYN Vulnerability Intelligence & Prioritization Engine provides continuous vulnerability aggregation, normalization, enrichment, correlation, prioritization, and remediation guidance across the enterprise. It transforms raw vulnerability data into explainable, risk-aware intelligence aligned with mission impact, threat activity, exploitability, and business context.

The engine enables proactive remediation by combining vulnerability intelligence, threat intelligence, attack surface management, asset criticality, predictive analytics, and executive governance.

It answers:

```
Which vulnerabilities present the highest organizational risk?
Which vulnerabilities are actively exploitable?
Which assets require immediate remediation?
What is the business and mission impact?
How should remediation be prioritized?
Which vulnerabilities are trending?
Can every prioritization decision be explained and audited?
```

4. Mission

The engine shall provide:

```
Vulnerability aggregation
Vulnerability normalization
Exploit intelligence correlation
Risk-based prioritization
Mission-aware prioritization
Remediation recommendations
Vulnerability trend analysis
Executive vulnerability summaries
Compliance reporting
Continuous reassessment
Vulnerability auditing
Policy-governed prioritization
```

5. Scope

5.1 In Scope

```
CVE ingestion
Scanner integration
Exploit intelligence
Asset criticality
Threat correlation
Mission impact
Risk scoring
Remediation prioritization
Exposure correlation
Executive reporting
```

5.2 Out of Scope

```
Patch deployment
Exploit execution
Penetration testing
Software inventory management
Application development lifecycle
```

6. Dependencies

IS-024 depends on:

- IS-001 AQELYN Kernel
- IS-002 Universal Object Model
- IS-003 AQELYN Event Bus
- IS-004 AQELYN Evidence Engine
- IS-005 AQELYN Knowledge Graph
- IS-006 AQELYN Trust Engine
- IS-007 AQELYN Mission Engine
- IS-008 AQELYN Workflow Engine
- IS-009 AQELYN Policy Engine
- IS-010 AQELYN Compliance & Governance Engine
- IS-011 Identity & Access Governance Engine
- IS-012 Asset & Configuration Governance Engine
- IS-013 Risk Intelligence Engine
- IS-014 Threat Intelligence Fusion Engine
- IS-015 Security Operations Engine
- IS-016 Digital Forensics Engine
- IS-017 Threat Detection & Analytics Engine
- IS-018 Automated Response & Orchestration Engine
- IS-019 Security Data Lake & Telemetry Platform
- IS-020 AI Decision Intelligence Engine
- IS-021 Predictive Analytics & Forecasting Engine
- IS-022 Executive Intelligence & Strategic Reporting Engine
- IS-023 Threat Exposure & Attack Surface Management Engine

7. High-Level Architecture

AQELYN Vulnerability Intelligence & Prioritization Engine

-
- Vulnerability Aggregation Engine
- Vulnerability Intelligence Engine
- Exploit Correlation Engine
- Prioritization Engine
- Remediation Recommendation Engine
- Trend Analysis Engine
- Compliance Correlation Engine
- Executive Reporting Connector
- Knowledge Graph Connector
- Data Lake Connector
- AI Decision Connector
- Event Publisher

8. Functional Requirements

FR-024-001 - Vulnerability Aggregation

The engine shall continuously aggregate:

- Scanner findings
- CVE feeds
- Vendor advisories
- Threat intelligence
- Cloud vulnerability data
- Container vulnerability data

FR-024-002 - Correlation

Correlate vulnerabilities with:

- Assets
- Threat intelligence
- Attack surface
- Mission criticality

Business impact
Exploit activity

FR-024-003 - Prioritization

Generate:

Risk score
Priority level
Mission impact
Business impact
Remediation priority
Executive summaries

FR-024-004 - Explainable Intelligence

Every prioritization decision shall include:

Evidence references
Confidence indicators
Risk rationale
Threat context
Asset lineage
Exploit intelligence

FR-024-005 - Governance

Support:

Approval workflows
Version control
Auditability
Policy validation
Executive review

FR-024-006 - Event Publication

Publish standardized events:

vulnerability.discovered
vulnerability.updated
vulnerability.prioritized
remediation.recommended
exploit.detected
risk.recalculated

9. Non-Functional Requirements

The engine shall provide:

Continuous assessment
Enterprise scalability
Low-latency prioritization
Explainability
Auditability
Repository stability
Backward compatibility

10. Core Vulnerability Lifecycle

Vulnerability Discovery
↓

```

Normalization
↓
Threat Correlation
↓
Risk Prioritization
↓
Remediation Recommendation
↓
Policy Validation
↓
Continuous Reassessment

```

11. Internal Component Architecture

The AQELYN Vulnerability Intelligence & Prioritization Engine is implemented as a modular vulnerability intelligence platform integrated with the AQELYN Kernel, Knowledge Graph, Security Data Lake, Risk Intelligence Engine, Threat Intelligence Fusion Engine, Attack Surface Management Engine, AI Decision Intelligence Engine, and Executive Intelligence Engine.

AQELYN Vulnerability Intelligence & Prioritization Engine

```

■
■■■ Vulnerability Aggregation Engine
■■■ Vulnerability Intelligence Engine
■■■ Exploit Correlation Engine
■■■ Prioritization Engine
■■■ Remediation Recommendation Engine
■■■ Trend Analysis Engine
■■■ Compliance Correlation Engine
■■■ Knowledge Graph Connector
■■■ Data Lake Connector
■■■ AI Decision Connector
■■■ Executive Reporting Connector
■■■ Event Publisher

```

12. Component Specifications

12.1 Vulnerability Aggregation Engine

Continuously aggregates vulnerability intelligence.

Capabilities:

```

Scanner ingestion
CVE ingestion
Vendor advisories
Cloud vulnerability feeds
Container vulnerability feeds
Third-party intelligence

```

12.2 Vulnerability Intelligence Engine

Maintains normalized vulnerability intelligence.

Supports:

```

CVE normalization
Severity normalization
Threat enrichment
Asset association
Historical tracking

```

12.3 Exploit Correlation Engine

Correlates vulnerabilities with exploit activity.

Produces:

- Known exploited vulnerabilities
- Threat actor correlation
- Exploit availability
- Exploit likelihood
- Threat campaigns

12.4 Prioritization Engine

Calculates enterprise remediation priorities.

Produces:

- Risk score
- Priority level
- Business impact
- Mission impact
- Remediation priority

12.5 Remediation Recommendation Engine

Generates remediation guidance.

Supports:

- Patch recommendations
- Mitigation guidance
- Compensating controls
- Risk acceptance options
- Executive summaries

12.6 Trend Analysis Engine

Analyzes vulnerability evolution.

Produces:

- Vulnerability trends
- Risk trends
- Remediation trends
- Exploit trends
- Exposure trends

12.7 Compliance Correlation Engine

Maps vulnerabilities to compliance obligations.

Supports:

- Regulatory controls
- Framework mappings
- Audit evidence
- Policy violations
- Compliance reporting

13. Universal Object Model Extensions

13.1 VulnerabilityRecord

```
VulnerabilityRecord:
vulnerability_id
cve_id
severity
risk_score
```

13.2 VulnerabilityAssessment

```
VulnerabilityAssessment:
assessment_id
confidence
recommendations
generated_at
```

13.3 RemediationPlan

```
RemediationPlan:
plan_id
priority
owner
target_date
```

13.4 ExploitCorrelation

```
ExploitCorrelation:
correlation_id
exploit_status
threat_context
confidence
```

14. Knowledge Graph Integration

Relationships:

```
Asset
↓
has
↓
Vulnerability
Vulnerability
↓
correlated_with
↓
Threat
Threat
↓
creates
↓
Risk
Risk
↓
drives
↓
Remediation
```

15. Event Bus Integration

15.1 Vulnerability Events

```
vulnerability.discovered
vulnerability.updated
vulnerability.closed
```

15.2 Prioritization Events

```
vulnerability.prioritized  
risk.recalculated
```

15.3 Exploit Events

```
exploit.detected  
exploit.updated
```

15.4 Remediation Events

```
remediation.recommended  
remediation.completed
```

16. Security Data Lake Integration

Consumes:

```
Scanner telemetry  
Historical vulnerability data  
Remediation history  
Asset telemetry  
Threat telemetry
```

17. AI Decision Intelligence Integration

Consumes:

```
Remediation recommendations  
Confidence scores  
Decision history  
Priority optimization  
Learning insights
```

18. Threat & Risk Intelligence Integration

Consumes:

```
Threat campaigns  
Exploit intelligence  
Risk scores  
Mission impact  
Business impact
```

19. Compliance Integration

Supports:

```
Compliance reporting  
Policy validation  
Regulatory mapping  
Audit evidence  
Governance reporting
```

20. Public APIs

20.1 Vulnerability API

```
GET /vulnerabilities
POST /vulnerabilities
GET /vulnerabilities/{id}
```

20.2 Assessment API

```
GET /assessments
POST /assessments
```

20.3 Remediation API

```
GET /remediation-plans
POST /remediation-plans
```

20.4 Intelligence API

```
GET /exploit-intelligence
GET /vulnerability-summary
```

21. Repository Impact

Implementation shall use the approved repository structure.

```
AQELYN/
  ■■■ src/
  ■ ■■■ vulnerability_intelligence/
  ■■■ tests/
  ■ ■■■ vulnerability_intelligence/
  ■■■ docs/
  ■ ■■■ vulnerability_intelligence/
  ■■■ api/
  ■ ■■■ vulnerability_intelligence/
  ■■■ archive/
```

No top-level repository modifications are permitted.

22. Security Architecture

The AQELYN Vulnerability Intelligence & Prioritization Engine is the trusted subsystem responsible for continuously aggregating, normalizing, enriching, prioritizing, and governing enterprise vulnerability intelligence.

Every vulnerability assessment shall be:

```
Explainable
Evidence-backed
Threat-aware
Mission-aware
Policy-governed
Risk-aware
Fully auditable
Continuously reassessed
```

22.1 Security Principles

```
Zero Trust
Defense in Depth
Least Privilege
```

Continuous Assessment
Explainable Intelligence
Policy Enforcement
Secure by Design
Continuous Validation

22.2 Authorization Model

Supported operational roles:

SOC Analyst
Vulnerability Analyst
Risk Analyst
Threat Intelligence Analyst
Mission Owner
Security Administrator
Compliance Officer
Executive Reviewer

All prioritization decisions shall be governed through the AQELYN Policy Engine.

22.3 Vulnerability Assessment Integrity

Vulnerability assessments shall maintain:

Unique assessment identifier
Vulnerability identifier
Asset references
Threat context
Exploit context
Risk score
Confidence indicators
Remediation guidance
Audit trail

Assessment history shall be append-only.

22.4 Remediation Integrity

Remediation plans shall maintain:

Plan identifier
Owner
Priority
Target date
Recommendation rationale
Risk reduction expectation
Approval state
Validation result
Audit record

No remediation recommendation shall be considered authoritative without supporting evidence and confidence records.

23. Vulnerability Lifecycle

23.1 Assessment Lifecycle

Vulnerability Discovery
↓
Normalization
↓
Threat Correlation
↓
Risk Prioritization
↓

```
Remediation Recommendation
↓
Continuous Reassessment
```

23.2 Remediation Lifecycle

```
Recommendation Generated
↓
Approval
↓
Remediation Initiated
↓
Validation
↓
Closure
```

23.3 Audit Lifecycle

```
Assessment Created
↓
Evidence Linked
↓
Policy Validated
↓
Audit Stored
```

24. Continuous Vulnerability Operations

The engine continuously evaluates:

```
New CVEs
Threat intelligence
Exploit activity
Asset criticality
Attack surface
Mission impact
Business risk
Remediation effectiveness
```

25. Performance Requirements

The engine shall support:

```
Continuous vulnerability ingestion
Low-latency prioritization
Enterprise-scale assessment
Concurrent remediation planning
High availability
Continuous operation
```

26. Scalability Requirements

The engine shall scale to support:

```
Global enterprises
Multi-cloud environments
Millions of vulnerability records
Large-scale scanner integration
Long-term historical intelligence
Distributed deployments
```

27. Audit Requirements

Every vulnerability operation shall generate immutable audit records.

Audit events include:

- Vulnerability discovered
- Assessment updated
- Risk recalculated
- Exploit correlated
- Remediation recommended
- Remediation completed

28. Failure Handling

28.1 Aggregation Failure

- Aggregation retried
- Failure recorded
- Administrator notified

28.2 Correlation Failure

- Threat correlation retried
- Fallback assessment generated
- Audit recorded

28.3 Prioritization Failure

- Risk recalculated
- Previous assessment retained
- Audit generated

28.4 Policy Failure

- Recommendation publication blocked
- Policy violation recorded
- Manual approval required

29. Testing Strategy

29.1 Unit Testing

Validate:

- Vulnerability Aggregation Engine
- Vulnerability Intelligence Engine
- Exploit Correlation Engine
- Prioritization Engine
- Remediation Recommendation Engine
- Compliance Correlation Engine

29.2 Integration Testing

Verify interaction with:

- Kernel
- Knowledge Graph
- Security Data Lake
- Threat Intelligence Engine

```
Risk Intelligence Engine
Attack Surface Management Engine
AI Decision Engine
Executive Reporting Engine
Compliance Engine
```

29.3 System Testing

Validate:

```
Vulnerability aggregation
Threat correlation
Risk prioritization
Remediation recommendations
Executive summaries
Audit generation
```

29.4 Security Testing

Verify:

```
Authorization
Policy enforcement
Explainability
Audit logging
Assessment integrity
```

29.5 Regression Testing

Verify IS-001 through IS-023 remain unaffected.

30. Acceptance Criteria

IS-024 is complete when:

```
Vulnerability Aggregation Engine implemented
Vulnerability Intelligence Engine implemented
Exploit Correlation Engine implemented
Prioritization Engine implemented
Remediation Recommendation Engine implemented
Repository unchanged
Testing documented
```

31. Repository Validation

Repository structure remains unchanged.

```
AQELYN/
■■■ src/vulnerability_intelligence/
■■■ tests/vulnerability_intelligence/
■■■ docs/vulnerability_intelligence/
■■■ api/vulnerability_intelligence/
■■■ archive/
```

No top-level repository modifications are permitted.

32. Engineering Summary

IS-024 introduces the AQELYN Vulnerability Intelligence & Prioritization Engine, providing enterprise-scale vulnerability aggregation, exploit correlation, mission-aware prioritization, remediation guidance, compliance correlation, explainable vulnerability intelligence, and policy-governed prioritization.

Major capabilities include:

- Vulnerability Aggregation
- Exploit Intelligence Correlation
- Risk-Based Prioritization
- Mission-Aware Prioritization
- Remediation Recommendations
- Compliance Correlation
- Executive Vulnerability Reporting
- Trend Analysis
- Continuous Reassessment
- Explainable Vulnerability Intelligence

The engine integrates with all previously completed AQELYN engines while preserving repository stability, modularity, and backward compatibility.

33. Specification Status

Specification ID : IS-024
Title : AQELYN Vulnerability Intelligence & Prioritization Engine
Status : COMPLETE
Engineering Archive : READY FOR GENERATION
Next Artifact : EA-0024

Engineering workflow status:

- EA-0001 COMPLETE
- EA-0002 COMPLETE
- EA-0003 COMPLETE
- EA-0004 COMPLETE
- EA-0005 COMPLETE
- EA-0006 COMPLETE
- EA-0007 COMPLETE
- EA-0008 COMPLETE
- EA-0009 COMPLETE
- EA-0010 COMPLETE
- EA-0011 COMPLETE
- EA-0012 COMPLETE
- EA-0013 COMPLETE
- EA-0014 COMPLETE
- EA-0015 COMPLETE
- EA-0016 COMPLETE
- EA-0017 COMPLETE
- EA-0018 COMPLETE
- EA-0019 COMPLETE
- EA-0020 COMPLETE
- EA-0021 COMPLETE
- EA-0022 COMPLETE
- EA-0023 COMPLETE
- IS-024 COMPLETE
- EA-0024 READY FOR GENERATION

34. EA-0024 Engineering Objective

The objective of IS-024 was to introduce a dedicated Vulnerability Intelligence & Prioritization Engine that enables AQELYN to continuously aggregate vulnerability findings, normalize vulnerability records, enrich them with threat and exploit intelligence, prioritize remediation using mission and risk context, and support explainable remediation decisions.

The engine extends AQELYN from exposure management into vulnerability intelligence and risk-based prioritization.

35. EA-0024 Engineering Summary

The implementation specification defines a modular subsystem responsible for:

- Vulnerability aggregation
- Vulnerability normalization
- Exploit intelligence correlation
- Risk-based prioritization
- Mission-aware prioritization
- Remediation recommendations
- Vulnerability trend analysis
- Compliance correlation
- Executive vulnerability summaries
- Knowledge Graph integration
- Security Data Lake integration
- Threat and Risk Intelligence integration
- AI Decision integration
- Event publishing

The engine integrates with all previously completed AQELYN engines while preserving architectural modularity.

36. Major Engineering Decisions

36.1 Decision 1 - Dedicated Vulnerability Intelligence Engine

Vulnerability intelligence and prioritization responsibilities are implemented as a standalone engine rather than embedded in Asset Governance, Attack Surface Management, or Risk Intelligence.

Rationale:

- Clear separation of vulnerability lifecycle from asset discovery and exposure management.
- Independent lifecycle and governance.
- Better support for exploit correlation and remediation prioritization.
- Improved traceability for vulnerability decisions.

36.2 Decision 2 - Prioritization Is Risk-Based, Not Severity-Only

Prioritization uses exploit intelligence, mission impact, asset criticality, exposure, threat activity, and business context.

Benefits:

- Critical remediation focuses on real organizational risk.
- CVSS or severity alone does not drive prioritization.
- Executive reporting aligns remediation with mission and business impact.

36.3 Decision 3 - Vulnerability Assessments Are Evidence-Backed

Every vulnerability assessment must reference evidence, asset lineage, exploit context, confidence, and remediation rationale.

Benefits:

- Prioritization becomes defensible.
- Audit and compliance reporting can review assessment logic.
- Remediation teams receive explainable guidance.

36.4 Decision 4 - Remediation Plans as First-Class Objects

Remediation plans are modeled as Universal Object Model extensions.

Benefits:

Remediation ownership becomes traceable.
 Target dates and validation status can be governed.
 Risk reduction can be measured after remediation.

36.5 Decision 5 - Event-Driven Vulnerability Lifecycle

Vulnerability discovery, updates, prioritization, exploit correlation, remediation recommendations, and closure events are published through the AQELYN Event Bus.

Examples include:

```
vulnerability.discovered
vulnerability.updated
vulnerability.closed
vulnerability.prioritized
risk.recalculated
exploit.detected
exploit.updated
remediation.recommended
remediation.completed
```

This maintains loose coupling between AQELYN engines.

36.6 Decision 6 - Universal Object Model Extension**New domain objects introduced include:**

```
VulnerabilityRecord
VulnerabilityAssessment
RemediationPlan
ExploitCorrelation
```

These extend the Universal Object Model without modifying existing object definitions.

37. Architectural Integration Summary

| Engine | Integration |

|---|---|

| IS-001 Kernel | Runtime lifecycle and service registration |

| IS-002 Universal Object Model | Vulnerability, assessment, remediation, exploit correlation objects |

| IS-003 Event Bus | Vulnerability, prioritization, exploit, remediation events |

| IS-004 Evidence Engine | Evidence references and vulnerability support |

| IS-005 Knowledge Graph | Asset, vulnerability, threat, risk, remediation relationships |

| IS-006 Trust Engine | Data confidence and assessment trust |

| IS-007 Mission Engine | Mission-aware prioritization |

| IS-008 Workflow Engine | Remediation and approval workflows |

| IS-009 Policy Engine | Prioritization governance, publication, validation policies |

| IS-010 Compliance Engine | Compliance reporting and regulatory mapping |

| IS-012 Asset Governance Engine | Asset ownership, criticality, classification |

| IS-013 Risk Intelligence Engine | Risk scoring and business impact |

| IS-014 Threat Intelligence Engine | Exploit intelligence, threat campaigns, actor context |

| IS-019 Security Data Lake | Scanner telemetry, history, vulnerability data |

| IS-020 AI Decision Engine | Remediation recommendations and learning insights |

| IS-022 Executive Reporting Engine | Executive vulnerability summaries |

| IS-023 Attack Surface Management Engine | Exposure correlation and attack surface context |

No existing engine required redesign.

38. Repository Impact Summary

Repository structure remains unchanged.

Implementation is expected within existing project directories, including:

```
AQELYN/  
■■■ src/vulnerability_intelligence/  
■■■ tests/vulnerability_intelligence/  
■■■ api/vulnerability_intelligence/  
■■■ docs/vulnerability_intelligence/  
■■■ archive/
```

No top-level directories were added, removed, or renamed.

39. Security Impact Summary

The specification introduces vulnerability-intelligence-specific security controls:

```
Policy-governed vulnerability prioritization  
Evidence-backed assessments  
Exploit-context correlation  
Mission-aware remediation priority  
Risk-based vulnerability scoring  
Remediation ownership tracking  
Assessment audit trail  
Compliance mapping  
Role-authorized vulnerability administration
```

No reduction in the security posture of existing components was identified.

40. Capabilities Added

The engine enables AQELYN to support:

```
Vulnerability aggregation  
Vulnerability normalization  
Exploit intelligence correlation  
Risk-based prioritization  
Mission-aware prioritization  
Remediation recommendations  
Vulnerability trend analysis  
Executive vulnerability summaries  
Compliance reporting  
Continuous reassessment  
Vulnerability auditing  
Policy-governed prioritization
```

41. Risks Identified

| Risk | Mitigation |

|---|---|

| Severity-only prioritization | Risk-based prioritization using mission, exploit, and exposure context |

Stale vulnerability data	Continuous ingestion and reassessment
False exploit correlation	Confidence indicators and threat context
Remediation overload	Prioritization and executive summaries
Poor ownership	RemediationPlan ownership and target dates
Unauthorized vulnerability visibility	Policy enforcement and role authorization
Compliance gaps	Compliance Correlation Engine
Weak remediation evidence	Evidence-backed assessment and validation

No critical architectural risks were identified that require redesign.

42. Verification Summary

The specification defines verification for:

Unit testing
Integration testing
System testing
Security testing
Regression testing

Acceptance criteria cover vulnerability aggregation, vulnerability intelligence, exploit correlation, prioritization, remediation recommendation, repository validation, and testing documentation.

43. Engineering Principles Confirmed

The implementation complies with established AQELYN principles:

Modular architecture
Event-driven communication
Immutable evidence references
Traceability
Explainability
Security by design
Repository stability
Backward compatibility
Governance-before-archive discipline

44. Dependencies

Required:

EA-0001 through EA-0023
IS-001 through IS-023

Enables:

IS-025 and subsequent asset discovery, inventory intelligence, cloud posture, identity exposure, vulnerability remediation, and cyber resilience components

45. Completion Record

Engineering Archive : EA-0024
Implementation Specification : IS-024
Title : AQELYN Vulnerability Intelligence & Prioritization Engine
Engineering Status : COMPLETE
Repository Status : UNCHANGED

```

Architecture Status : EXTENDED
Backward Compatibility : MAINTAINED
Engineering Rule :
IS Completed
↓
EA Generated
↓
Continue

```

46. Archive Index Update

```

EA-0001 IS-001 AQELYN Kernel
EA-0002 IS-002 Universal Object Model
EA-0003 IS-003 AQELYN Event Bus
EA-0004 IS-004 AQELYN Evidence Engine
EA-0005 IS-005 AQELYN Knowledge Graph
EA-0006 IS-006 AQELYN Trust Engine
EA-0007 IS-007 AQELYN Mission Engine
EA-0008 IS-008 AQELYN Workflow Engine
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EA-0010 IS-010 AQELYN Compliance & Governance Engine
EA-0011 IS-011 AQELYN Identity & Access Governance Engine
EA-0012 IS-012 AQELYN Asset & Configuration Governance Engine
EA-0013 IS-013 AQELYN Risk Intelligence Engine
EA-0014 IS-014 AQELYN Threat Intelligence Fusion Engine
EA-0015 IS-015 AQELYN Security Operations (SOC) Engine
EA-0016 IS-016 AQELYN Digital Forensics Engine
EA-0017 IS-017 AQELYN Threat Detection & Analytics Engine
EA-0018 IS-018 AQELYN Automated Response & Orchestration Engine
EA-0019 IS-019 AQELYN Security Data Lake & Telemetry Platform
EA-0020 IS-020 AQELYN AI Decision Intelligence Engine
EA-0021 IS-021 AQELYN Predictive Analytics & Forecasting Engine
EA-0022 IS-022 AQELYN Executive Intelligence & Strategic Reporting Engine
EA-0023 IS-023 AQELYN Threat Exposure & Attack Surface Management Engine
EA-0024 IS-024 AQELYN Vulnerability Intelligence & Prioritization Engine

```

47. Engineering Phase Status

Completed Engineering Archives : EA-0001 through EA-0024

Current Status:
EA-0024 COMPLETE

Next Implementation Specification:
IS-025 - AQELYN Cyber Asset Discovery & Inventory Intelligence Engine

EA-0024 is completed and archived. The engineering workflow is consistent with the project rule:

Implementation Specification -> Engineering Archive -> Continue

From this point onward, the next engineering artifact is IS-025.

48. Engineering Archive Publication Standard

EA-0024 follows the AQELYN Engineering Archive Publication Standard.

Required package structure:

```

EA-xxxx/
■
■■■ diagrams/
■ ■■■ Architecture.svg
■ ■■■ Component.svg
■ ■■■ Workflow.svg
■ ■■■ EventFlow.svg

```

```
■ ■■■ Integration.svg
■
■ ■■■ examples/
■ ■■■ example_*.md
■
■ ■■■ HTML/
■ ■■■ index.html
■ ■■■ styles.css
■ ■■■ assets/
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■ ■■■ images/
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■ ■■■ journal/
■ ■■■ Engineering_Journal.md
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■ ■■■ manifest/
■ ■■■ manifest.json
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■ ■■■ MD/
■ ■■■ EA-xxxx.md
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■ ■■■ PDF/
■ ■■■ EA-xxxx.pdf
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■ ■■■ requirements/
■ ■■■ Requirements_Matrix.md
■
■ ■■■ traceability/
■ ■■■ Traceability_Matrix.md
■
■ ■■■ README.md
```

The master Markdown is the source of truth. The PDF and HTML are generated from the same master Markdown and must not omit sections.

49. Requirements Matrix

Requirement ID	Requirement	Evidence in Archive	Status
FR-024-001	Aggregate vulnerability intelligence	Sections 8, 12, 23	Complete
FR-024-002	Correlate vulnerabilities	Sections 8, 12, 18	Complete
FR-024-003	Prioritize vulnerabilities	Sections 8, 12, 23	Complete
FR-024-004	Provide explainable intelligence	Sections 8, 22, 36	Complete
FR-024-005	Support governance	Sections 8, 22, 23	Complete
FR-024-006	Publish vulnerability events	Sections 8, 15, 36	Complete
NFR-024-001	Continuous assessment	Sections 9, 24	Complete
NFR-024-002	Enterprise scalability	Sections 9, 25, 26	Complete
NFR-024-003	Low-latency prioritization	Sections 9, 25	Complete
NFR-024-004	Explainability	Sections 9, 22, 36	Complete
NFR-024-005	Auditability	Sections 9, 27, 39	Complete
NFR-024-006	Repository stability	Sections 21, 31, 38	Complete

50. Traceability Matrix

Source	Target	Relationship
IS-024 Purpose	EA-0024 Objective	Defines why the engine exists
Vulnerability Aggregation Engine	FR-024-001	Aggregates vulnerability data

| Vulnerability Intelligence Engine | FR-024-001 | Normalizes and enriches vulnerabilities |

| Exploit Correlation Engine | FR-024-002 | Correlates exploit activity |

| Prioritization Engine | FR-024-003 | Calculates remediation priority |

| Remediation Recommendation Engine | Remediation lifecycle | Generates remediation guidance |

| Compliance Correlation Engine | Governance and reporting | Maps vulnerabilities to compliance |

| Event Publisher | FR-024-006 | Publishes vulnerability events |

| Security Data Lake Integration | Historical vulnerability data | Supplies telemetry and history |

| AI Decision Integration | Remediation recommendations | Supplies confidence and decision support |

| Threat & Risk Intelligence Integration | Threat-aware prioritization | Supplies exploit and risk context |

| Compliance Integration | Audit and regulatory mapping | Supports reporting and evidence |

| Repository Validation | Repository Standard | Confirms no top-level redesign |

51. Engineering Journal

Journal Entry - EA-0024

EA-0024 was created to archive completion of IS-024 - AQELYN Vulnerability Intelligence & Prioritization Engine.

The archive records the expansion of AQELYN into vulnerability intelligence and risk-based remediation prioritization. IS-024 defines the structure needed to aggregate vulnerability findings, normalize CVE and scanner data, correlate exploit intelligence, calculate risk-based priority, generate remediation recommendations, analyze trends, map compliance requirements, and publish vulnerability events.

The engineering design preserves the fixed AQELYN repository structure and maintains backward compatibility with previously completed engines.

Lessons Learned

Vulnerability intelligence must be modeled separately from asset discovery and exposure management. Asset discovery identifies what exists, exposure management identifies what is reachable, and vulnerability intelligence identifies what is weak and how remediation should be prioritized.

Governance Note

EA-0024 follows the master-document publication workflow. The Markdown file is the authoritative source, and PDF/HTML representations are generated from the same content.

52. Examples

52.1 Example Vulnerability Record

```
vulnerability_id: VULN-0001
cve_id: CVE-2026-0001
severity: critical
risk_score: 94
asset_id: ASSET-1001
exploit_status: known_exploited
```

52.2 Example Vulnerability Assessment

```
assessment_id: VASSESS-1001
confidence: 0.92
```

```
recommendations:  
- apply vendor patch  
- isolate affected public service until remediation  
- validate compensating controls  
generated_at: 2026-07-07T12:00:00Z
```

52.3 Example Remediation Plan

```
plan_id: REM-2001  
priority: immediate  
owner: infrastructure_security_team  
target_date: 2026-07-10  
validation_required: true
```

52.4 Example Vulnerability Event

```
{  
  "event_type": "vulnerability.prioritized",  
  "vulnerability_id": "VULN-0001",  
  "risk_score": 94,  
  "priority": "immediate",  
  "source_engine": "aqelyn_vulnerability_intelligence_prioritization_engine"  
}
```

53. Manifest Summary

Archive contents include:

```
README.md  
MD/EA-0024.md  
PDF/EA-0024.pdf  
HTML/index.html  
HTML/styles.css  
manifest/manifest.json  
requirements/Requirements_Matrix.md  
traceability/Traceability_Matrix.md  
journal/Engineering_Journal.md  
diagrams/Architecture.svg  
diagrams/Component.svg  
diagrams/Workflow.svg  
diagrams/EventFlow.svg  
diagrams/Integration.svg  
examples/example_vulnerability_intelligence.md
```

54. Final Archive Statement

EA-0024 is the Engineering Archive for IS-024 - AQELYN Vulnerability Intelligence & Prioritization Engine.

It records the completed specification, the architectural decisions, the integration model, the repository impact, the risk posture, verification requirements, acceptance criteria, archive index update, and the engineering publication standard.

```
EA-0024 COMPLETE  
IS-024 COMPLETE  
NEXT: IS-025
```